SujetStore

Educational Archive Platform for the Algerian Curriculum

Technical Manual — Architecture, Features & Roadmap

Version 2.0 — March 2026

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1. Introduction: Rethinking Digital Learning in Algeria

1.1. The Problem

In the current Algerian educational landscape, students face a fragmented digital experience. Exam papers, quizzes, and study materials are scattered across social media groups, unmoderated forums, and low-quality scan repositories. Key issues include:

- **Content Dispersion:** Thousands of documents are posted daily across Facebook groups and Telegram channels with no indexing, versioning, or quality control. Finding a specific exam from a specific trimester becomes an exercise in frustration.
- **No Pedagogical Structure:** Materials are dumped without classification by trimester, type, difficulty, or alignment with the official Ministry of Education curriculum.
- **Poor Visual Quality:** Most documents are low-resolution phone scans with inconsistent fonts, skewed pages, and illegible handwriting.
- **Hostile User Experience:** Existing platforms (e.g., DzExams) rely on aggressive advertising, intrusive pop-ups, and confusing navigation that actively hinders the learning process.

1.2. The Vision

SujetStore was conceived as a purpose-built, open-source educational archive platform that directly addresses these pain points. It provides a clean, focused, and fast experience for browsing, filtering, and reading exam papers and study materials for the Algerian curriculum — from primary school through high school.

1.3. Core Objectives

- **Absolute Focus:** Deliver content within a distraction-free, ad-free interface designed for deep study and review. No pop-ups, no banners, no autoplay videos.
- **Hierarchical Archiving:** Organize the entire curriculum into a strict, navigable taxonomy: *Level* → *Year* → *Subject* → *Trimester* → *Document Type*.
- **Empowering Content Creators:** Provide a powerful yet simple admin CMS that allows educators to upload, categorize, and manage documents without any technical knowledge or database interaction.
- **Resilient Infrastructure:** Leverage a modern, lightweight stack (SvelteKit + Bun + SQLite) that can scale to millions of concurrent users without infrastructure collapse.

2. Pedagogical Strategy: Quality Over Quantity

The platform adopts a philosophy centered on content quality and pedagogical alignment with the official curricula issued by the Algerian Ministry of National Education. Successful digital education is not about digitizing textbooks or flooding students with files — it is about curating the right material, in the right order, with the right structure.

2.1. Competency-Based Organization

Documents on SujetStore are not random compilations. Every uploaded quiz, exam, or summary is intentionally placed within a pedagogical context:

- **Curriculum Alignment:** Each document is tagged with its trimester, type (quiz, exam, lesson/summary), source (official or proposed), and academic year. This ensures students can filter down to exactly what they need for their current revision focus.
- **Progressive Difficulty:** The archival structure naturally supports progression — from Term 1 (foundational concepts) through Term 3 (synthesis and integration activities), mirroring the official academic calendar.
- **Trimester-Based Navigation:** The Segmented Control tab interface explicitly separates content by trimester, preventing the cognitive overload that comes from seeing an undifferentiated list of dozens of documents.

2.2. Fighting Content Duplication

The platform actively avoids the "repeat and hope" approach common on social media groups, where the same exam gets re-posted dozens of times with no context. When similar exercises target the same learning objective, the system makes this relationship explicit — reinforcing the pedagogical intent rather than creating noise.

2.3. Visual Consistency with Typst

To eliminate the visual fatigue caused by low-quality scans and inconsistent formatting, the platform supports integration with the `Typst` typesetting system for generating high-quality, professionally formatted PDF documents.

- **Custom Templates:** Purpose-built templates for each subject area. Mathematics documents handle complex equations and graphs using Typst's native rendering. Literary subjects emphasize proper Arabic typography and diacritical marks.
- **Uniform Branding:** All platform-generated documents share a consistent visual identity — clean margins, professional headers, and legible fonts — creating a stark contrast with the typical scan quality found elsewhere.

3. Features & User Experience

The platform was designed with the principle that *"simplicity creates effectiveness."* Every feature serves a clear purpose: helping students find, filter, and study documents as quickly as possible.

3.1. Student-Facing Interface

3.1.1. Homepage & Navigation Hierarchy

The homepage presents the three educational levels of the Algerian system as prominent, interactive cards:

- **Primary** — Grades 1 through 5
- **Middle School** — Grades 1AM through 4AM
- **Secondary** — Grades 1AS through 3AS (across all streams)

Each card displays live statistics (number of years and documents available) and links directly to the year selection page. A real-time stats bar shows the total count of documents, exams, quizzes, and subjects across the entire platform.

A global breadcrumb navigation system (`Level / Year / Subject`) ensures the student always knows their location and can navigate back to any level with a single click.

Figure 1: The SujetStore homepage with level cards, stats bar, and feature highlights

3.1.2. Mobile-First Responsive Design

The entire interface is built with a **mobile-first** approach, recognizing that the majority of Algerian students access educational content from smartphones. Key responsive behaviors include:

- Filter controls stack vertically on small screens and align horizontally on desktop.
- Tab navigation scrolls horizontally when screen width is constrained.
- Document rows adapt their layout for optimal touch interaction.
- All interactive elements meet minimum touch target sizes (44×44px).

3.1.3. Dark & Light Mode with Shadcn-Svelte Theming

The platform implements a complete dual-theme system using `Shadcn-Svelte` semantic color tokens. Every component — from buttons and dropdowns to cards and toggle switches — automatically adapts to the user's preferred color scheme.

Key tokens used throughout the interface:

- `bg-background` / `text-foreground` — Base page colors
- `bg-card` / `text-card-foreground` — Elevated surface containers
- `bg-muted` / `text-muted-foreground` — Subdued backgrounds and secondary text
- `bg-primary` / `text-primary-foreground` — Active and accent elements
- `bg-accent` / `text-accent-foreground` — Hover and interactive states

This approach guarantees that **no hardcoded colors** appear in the codebase, ensuring perfect contrast in both light and dark modes.

3.1.4. Native RTL & Arabic Typography

The design is built from the ground up for Right-to-Left (RTL) text direction, with the `Cairo` font applied globally for optimal Arabic readability. Flex layouts, padding, margins, and icon positions are all RTL-aware.

3.2. Trimester Tabs — Segmented Control

The subject page organizes documents using a **Segmented Control** interface — a horizontal tab bar centered on the page with three segments: *Term 1*, *Term 2*, and *Term 3*.

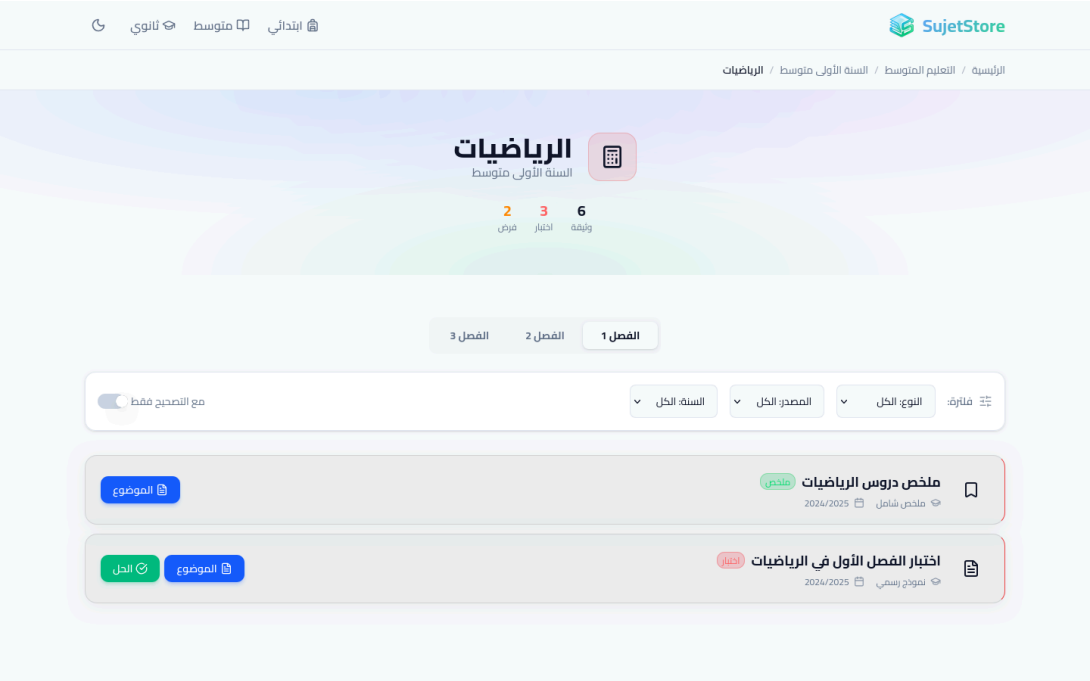

Figure 2: Subject page showing Segmented Control tabs, contextual filter bar, and document list

Implementation details:

- Built with a `bg-muted` container and `bg-background shadow-sm` for the active segment, following the standard Shadcn UI `TabsList` / `TabsTrigger` pattern.
- **Smart Auto-Selection:** On page load, the system automatically selects the first trimester that contains documents, preventing the user from landing on an empty tab.
- Documents with `trimester_id = null` (e.g., lessons and summaries) are displayed across **all** trimester tabs, ensuring they are always accessible regardless of the active tab.

3.3. Contextual Filter Bar

Directly below the trimester tabs, a horizontal Filter Bar allows fine-grained filtering of documents:

Filter	Type	Options
Document Type	Dropdown	All · Quiz · Exam · Lesson/Summary
Source	Dropdown	All · Official · Proposed
Academic Year	Dropdown	All · 2024 · 2023 · 2022
Has Solution	Toggle Switch	On/Off — filters to documents with corrections only

All filtering is performed **client-side and reactively** using Svelte 5's `$derived.by()` rune, providing instant results with zero network requests. When no documents match the current filter combination, a clear empty-state message with a filter icon is displayed.

The filter bar uses Shadcn-compatible classes (`bg-card`, `bg-background`, `text-foreground`) and adapts its layout based on screen size:

- **Mobile:** Controls stack vertically, each taking full width
- **Desktop:** Controls align horizontally in a single row

3.4. Embedded PDF Document Viewer

When a student clicks a document row, a fullscreen modal opens with an embedded PDF viewer featuring:

- **Tabbed Navigation:** Separate "Subject" and "Solution" tabs for switching between the exam paper and its model answer.
- **Direct Download:** A download button (📥) for saving the PDF locally.
- **Background Printing:** A print button (🖨) that triggers printing without opening disruptive browser dialogs.
- **Visited Tracking:** Documents the student has already opened are marked with a checkmark (✓) and a visual color change on the button, powered by `localStorage`.

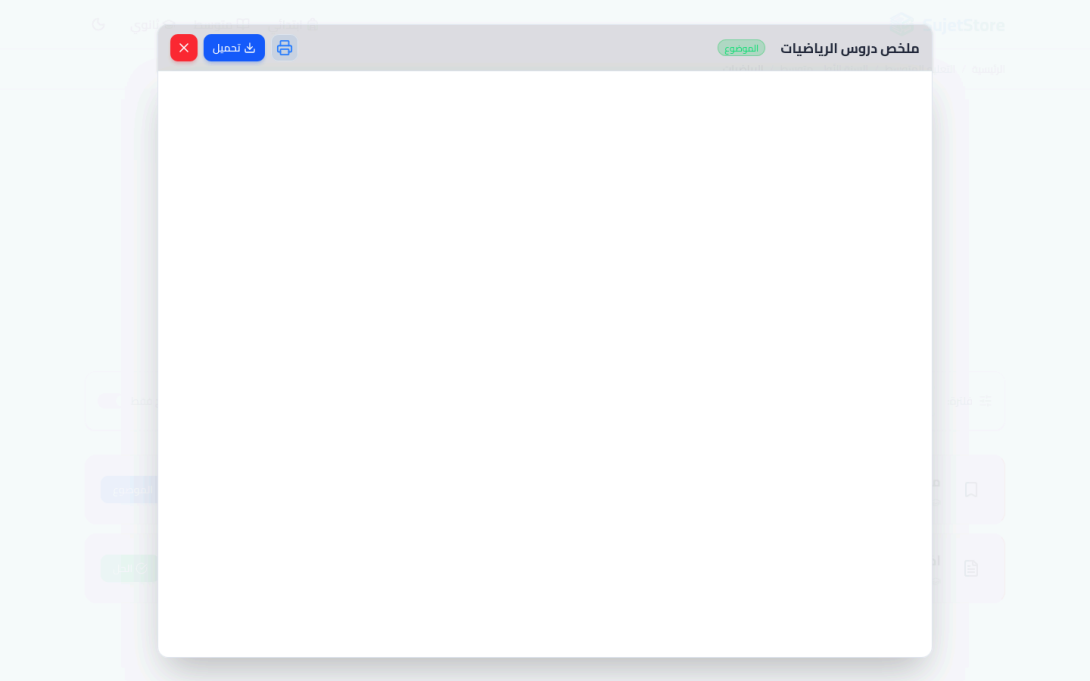

Figure 3: Fullscreen PDF viewer with Subject/Solution tab switching, download, and print controls

3.5. Smart Local Tracking

To respect student privacy while still providing useful study progress features, the platform implements a **localStorage-based tracking system**:

- Documents are marked as "viewed" when opened, and this state persists across browser sessions.
- No server-side tracking, no user accounts required, and no data is transmitted to external services.
- Subject and Solution buttons visually reflect their visited state, enabling students to quickly identify which materials they have already reviewed.

3.6. Admin Content Management System (CMS)

The platform includes a full-featured admin panel protected by session-based authentication using `bcryptjs` password hashing.

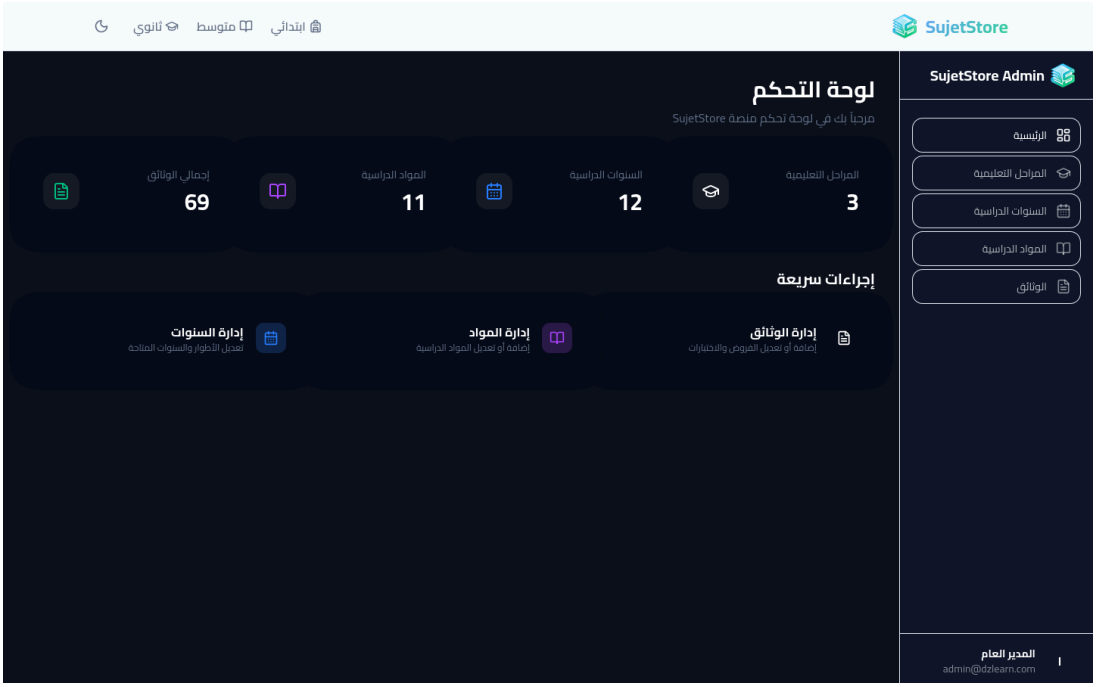

Figure 4: Admin dashboard with statistics, quick actions, and sidebar navigation

3.6.1. Admin Capabilities

- **Hierarchical CRUD:** Full create, read, update, and delete operations for educational levels, academic years, subjects, and documents.
- **Modal-Based Editing:** All add/edit operations use floating modals for fast interaction without full page navigation.
- **Confirmation Dialogs:** Destructive operations (e.g., deleting a document) require explicit confirmation through a custom `ConfirmModal` component, preventing accidental data loss.

- **File Upload & Linking:** Documents can be uploaded directly or linked via URL, with preview capabilities before publishing.
- **Mobile-Friendly Dashboard:** The admin interface is fully responsive, allowing content management from any device.

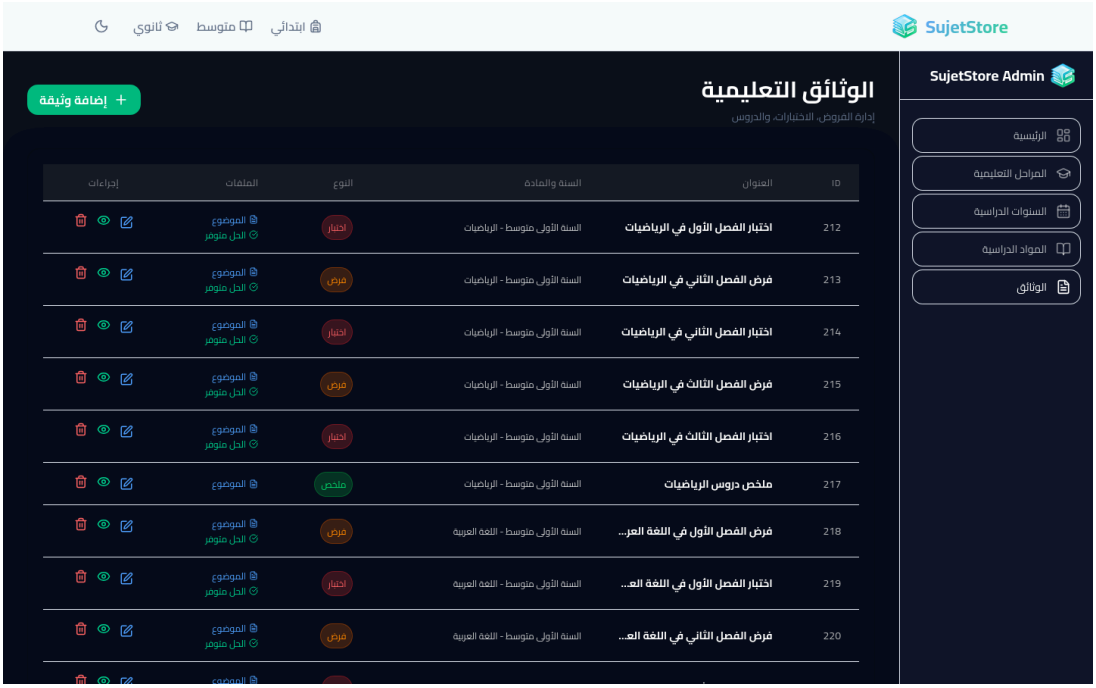

Figure 5: Document management table with type badges, file links, and CRUD actions

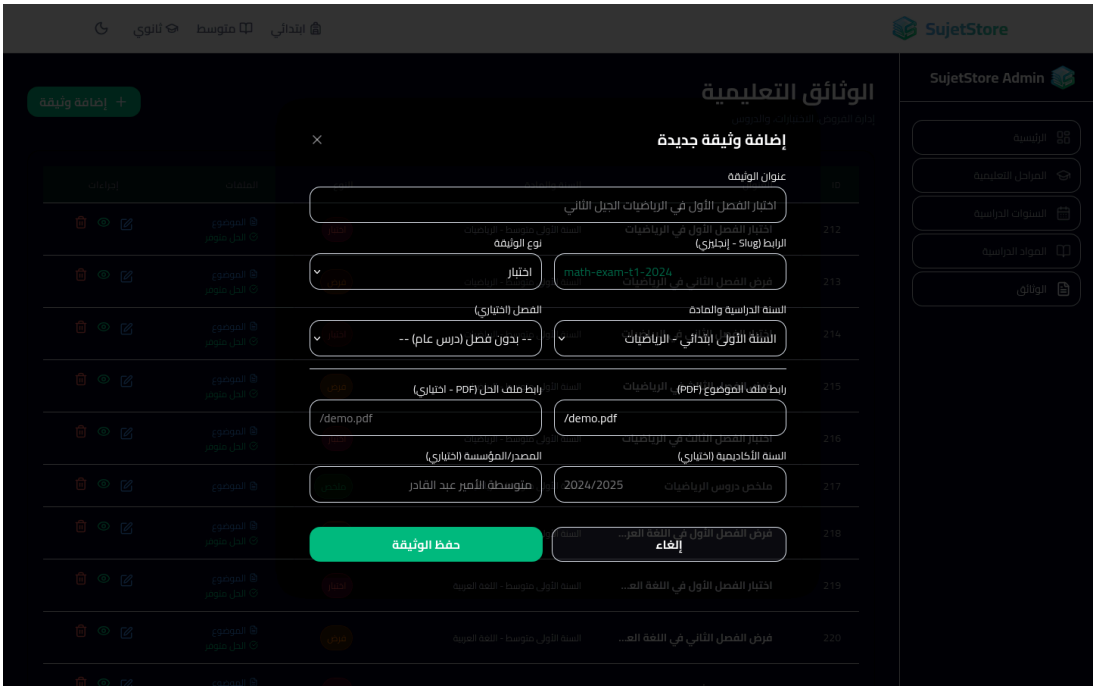

Figure 6: Floating modal for adding a new document with form fields and dropdowns

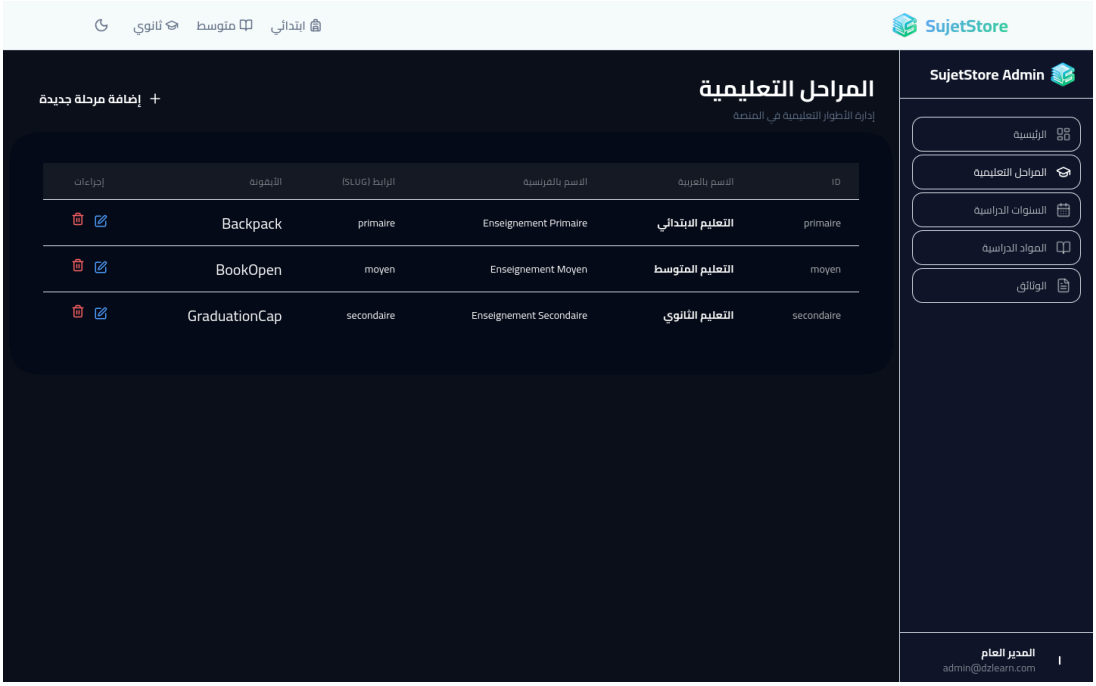

Figure 7: Educational levels management with icons, slugs, and edit/delete actions

4. Technical Architecture & Developer Guide

SujetStore is built on a modern, performance-oriented stack carefully selected for speed, simplicity, and developer experience.

4.1. The Tech Stack

Component	Technology	Rationale
Runtime	Bun (v1.1+)	3× faster than Node.js. Built-in bundler, test runner, and native SQLite driver.
Framework	SvelteKit (Svelte 5)	Full-stack framework with SSR. No Virtual DOM — minimal client-side overhead.
Database	SQLite	Embedded, zero-config, read-optimized. Uses `bun:sqlite` for native performance.
ORM	Drizzle ORM	Lightweight, type-safe query builder with full TypeScript support.
CSS	TailwindCSS v4	Utility-first CSS with native CSS variable support and zero-config setup.
UI Components	Shadcn-Svelte	Accessible, themeable component library with semantic color tokens.
Auth	Custom (bcryptjs)	Session-based authentication stored in SQLite. No external dependencies.

Deployment	Caddy + systemd	Auto-SSL reverse proxy with process management.

4.2. Architecture Philosophy: The Compact Monolith

Rather than adopting microservices or a headless frontend/backend split, SujetStore uses a **Compact Monolith** architecture:

- **Single Process:** The frontend (Svelte 5) and backend (SvelteKit server routes) run within a single Bun process.
- **File-Based Routing:** Routes are defined by the filesystem structure under `src/routes/`, eliminating the need for manual route configuration.
- **Co-located Data:** Both SQLite databases (`content.db` and `users.db`) live on the same server, accessed via `bun:sqlite` for sub-millisecond query performance.

4.3. Dual-Database Strategy

The system uses two separate SQLite databases to cleanly separate concerns:

4.3.1. Content Database (`content.db`) — Read-Only in Production

Stores all educational content: levels, years, subjects, trimesters, and documents. This database is generated offline using the seed script and deployed as a single file. Updates are performed by replacing the entire file.

4.3.2. Users Database (`users.db`) — Read-Write

Stores admin user accounts and session data. WAL mode is enabled for concurrent read/write access:

```
PRAGMA journal_mode = WAL;
PRAGMA busy_timeout = 5000;
PRAGMA synchronous = NORMAL;
```

This is the only database that requires periodic backup.

4.4. Project Structure

```
sujetstore/
├── src/
│   ├── lib/
│   │   ├── components/           # Shared Svelte components
│   │   │   ├── ConfirmModal.svelte   # Destructive action confirmation
│   │   │   └── DynamicIcon.svelte    # Runtime icon rendering
│   │   └── server/
│   │       └── db/               # Drizzle setup & schemas
```

```
│   │           ├── index.ts          # DB connection (bun:sqlite)
│   │           ├── schema-content.ts # Content tables
│   │           └── schema-users.ts   # Auth tables
│   ├── routes/
│   │   ├── +page.svelte              # Homepage (level cards + stats)
│   │   ├── [level]/
│   │   │   ├── +page.svelte          # Year grid for a level
│   │   │   └── [year]/
│   │   │       ├── +page.svelte      # Subject grid for a year
│   │   │       └── [subject]/
│   │   │           ├── +page.svelte  # Document list (tabs + filters)
│   │   │           └── [docSlug]/
│   │   │               └── +page.svelte  # PDF viewer
│   │   └── admin/
│   │       ├── +page.svelte          # Admin dashboard
│   │       ├── login/                # Authentication
│   │       ├── levels/               # Level management
│   │       ├── years/                # Year management
│   │       ├── subjects/             # Subject management
│   │       └── documents/            # Document management
│   ├── hooks.server.ts               # Auth middleware
│   └── app.html & app.css            # Root layout & global styles
├── book/                             # This manual (Typst source)
├── data/
│   ├── content.db                    # Educational content
│   └── users.db                      # Admin accounts & sessions
├── scripts/
│   └── seed-education.ts             # Database seeding script
├── deploy.sh                         # Deployment automation
└── ARCHITECTURE.md                   # Foundational architecture doc
```

4.5. Key Design Patterns

4.5.1. Reactive Filtering with Svelte 5 Runes

Document filtering uses Svelte 5's `$derived.by()` rune for zero-latency client-side filtering:

```
let filteredDocs = $derived.by(() => {
    let docs = data.documents.filter(
        (d) => d.trimester_id === activeTrimesterTab
            || d.trimester_id === null
```

```
    );
    if (filterType !== 'all') docs = docs.filter(...);
    if (filterSource !== 'all') docs = docs.filter(...);
    // ... additional filters
    return docs;
});
```

4.5.2. Auto-Tab Selection

On page mount, the system scans trimesters in order and automatically activates the first one with available documents:

```
onMount(() => {
    const firstTrimester = data.trimesters.find(
        (t) => getDocsByTrimester(t.id).length > 0
    );
    if (firstTrimester) activeTrimesterTab = firstTrimester.id;
});
```

4.5.3. Local State Persistence

Visited document tracking uses `localStorage` with `Set`-based serialization:

```
const stored = localStorage.getItem('visited_docs');
visitedDocs = new Set(JSON.parse(stored));
// On update:
localStorage.setItem('visited_docs',
    JSON.stringify([...visitedDocs]));
```

4.6. Development Workflow

```
# Install dependencies
bun install

# Seed the database with sample data
bun run scripts/seed-education.ts

# Start the development server
bun run dev

# Type-check the project
bunx svelte-check
```

5. Future Roadmap & Business Model

The initial release of SujetStore is a foundation — not a final product. The platform is designed with clear expansion vectors that build on the existing architecture.

5.1. Phase 1: Interactive Assessment Engine (Quiz System)

Moving beyond the traditional "solve on paper, compare with answer key" approach, the platform will integrate an interactive exercise engine:

- **Multiple Question Types:** MCQ (multiple choice), term matching, fill-in-the-blank, and drag-and-drop ordering exercises.
- **Instant Feedback:** Real-time scoring with targeted hints when a student answers incorrectly, identifying the specific concept that needs review.
- **Curriculum-Aligned:** Quizzes mapped directly to the competencies defined in the official Ministry of Education curriculum for each subject and trimester.

5.2. Phase 2: Video Solutions & Streaming Architecture

To support deeper understanding — especially for subjects like Physics and Mathematics where static PDFs have limitations — the platform will integrate video walkthroughs using a "pen and paper" recording style that mimics a teacher sitting beside the student.

The streaming infrastructure is designed for cost-efficiency at scale:

Technology	Purpose
HLS Protocol	Adaptive bitrate streaming with 10-second chunks. Instant playback without full download.
AES-128 Encryption	Per-chunk encryption to prevent unauthorized download via browser extensions.
Backblaze B2	Cold storage at 1/5th the cost of Amazon S3.
Cloudflare CDN	Zero egress fees via the Bandwidth Alliance. Edge distribution across North Africa and Europe.

This architecture enables the platform to serve millions of video views while paying only for storage — not bandwidth.

5.3. Phase 3: Centralized Analytics Dashboard

Transitioning from local tracking to a server-side analytics system:

- **Student Progress Profiles:** Visual progress indicators showing completed vs. remaining documents as percentages per subject and trimester.

- **Automated Weakness Detection:** Analysis of quiz errors to identify knowledge gaps and generate targeted review recommendations.
- **Parent Dashboards:** Summary statistics showing study effort and progress by trimester and academic year.

5.4. Business Model: Freemium with Zero Advertising

SujetStore adopts a **Product-Led Growth** (PLG) strategy with an absolute zero-advertising policy. This is a strategic decision to build trust with students and parents, and to maintain a distraction-free study environment.

5.4.1. Revenue Structure

Tier	Details
Free (Forever)	Full access to the document archive, PDF viewer, Typst-formatted documents, filters, and local tracking. This tier drives organic traffic.
Premium	Interactive quizzes, video solutions, and advanced analytics behind a paywall.

5.4.2. Pricing Strategy & Unit Economics

The pricing model is designed to be accessible to all socioeconomic segments of Algerian society:

- **Per-Subject:** 100 DZD ($0.70 USD) per subject for a full academic year.
- **Full-Level Bundle (Decoy Effect):** 1,000 DZD ($7 USD) for all subjects within a level — effectively providing 2 free subjects and driving the majority of users toward the bundle.
- **Market Penetration Target:** With approximately 11 million students in the Algerian educational system, reaching less than 1% (≈100,000 subscribers) at the bundle price would generate 100 million DZD in sustainable annual revenue — sufficient to compensate content creators and fund continuous platform development.

5.5. Deployment & Operations

5.5.1. Current Infrastructure

Component	Specification
Provider	Contabo VPS
Resources	12 vCPU · 48GB RAM · NVMe Storage
Web Server	Caddy (auto-SSL reverse proxy)
Process Manager	systemd service unit

Backup Strategy	Daily SQLite `.backup` of `users.db` with 7-day retention

5.5.2. Deployment Commands

```
./deploy.sh full    # Initial setup + deployment
./deploy.sh deploy  # Subsequent updates
```

5.5.3. Security Layers

Layer	Implementation
Transport	HTTPS enforced via Caddy auto-SSL
Cookies	`HttpOnly` · `Secure` · `SameSite=Lax`
Passwords	bcryptjs hashing
SQL Injection	Prepared statements via Drizzle ORM
Rate Limiting	Request throttling on `/admin/login` endpoints